Dirac equation 1

Consider the Schrodinger equation

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x},t) = H\psi(\mathbf{x},t)$$

The Dirac equation has for the Hamiltonian H

$$H = \alpha_x p_x c + \alpha_y p_y c + \alpha_z p_z c + \beta mc^2$$

where α and β are 4×4 matrices

$$\alpha_x = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \qquad \alpha_y = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} \qquad \alpha_z = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

and

$$\beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Substitute H into the Schrodinger equation to obtain

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x},t) = \left(\alpha_x p_x c + \alpha_y p_y c + \alpha_z p_z c + \beta mc^2\right)\psi(\mathbf{x},t)$$

Rearrange as

$$\left(i\hbar\frac{\partial}{\partial t} - \alpha_x p_x c - \alpha_y p_y c - \alpha_z p_z c\right)\psi(\mathbf{x},t) = \beta mc^2\psi(\mathbf{x},t)$$

Left-multiply by β.

$$\left(i\hbar\beta\frac{\partial}{\partial t} - \beta\alpha_x p_x c - \beta\alpha_y p_y c - \beta\alpha_z p_x c\right)\psi(\mathbf{x},t) = mc^2\psi(\mathbf{x},t)$$

Define γ matrices

$$\gamma_t = \beta \qquad \gamma_x = \beta\alpha_x \qquad \gamma_y = \beta\alpha_y \qquad \gamma_z = \beta\alpha_z$$

and write

$$\left(i\hbar\gamma_t\frac{\partial}{\partial t} - \gamma_x p_x c - \gamma_y p_y c - \gamma_z p_x c\right)\psi(\mathbf{x},t) = mc^2\psi(\mathbf{x},t)$$

Substitute $\mathbf{p} = -i\hbar\nabla$ for the momentum operators to obtain

$$i\hbar\left(\gamma_t\frac{\partial}{\partial t} + c\gamma_x\frac{\partial}{\partial x} + c\gamma_y\frac{\partial}{\partial y} + c\gamma_z\frac{\partial}{\partial z}\right)\psi(\mathbf{x},t) = mc^2\psi(\mathbf{x},t)$$

In spacetime notation

$$i\hbar\left(\gamma^0\frac{\partial}{\partial x^0}+c\gamma^1\frac{\partial}{\partial x^1}+c\gamma^2\frac{\partial}{\partial x^2}+c\gamma^3\frac{\partial}{\partial x^3}\right)\psi(x^\mu)=mc^2\psi(x^\mu)$$

Verify dimensions.

$$\begin{aligned}\hbar\frac{\partial}{\partial t}&=[\mathrm{J\,s}]\,[\mathrm{s}^{-1}]=[\mathrm{J}]\\ \hbar c\frac{\partial}{\partial x}&=[\mathrm{J\,s}]\,[\mathrm{m\,s}^{-1}]\,[\mathrm{m}^{-1}]=[\mathrm{J}]\\ mc^2&=[\mathrm{J}]\end{aligned}$$

Eigenmath script